

## 2026 九 年 级 数 学 5 月 适 应 性 训 练 参 考 答 案

一、选择题（共 10 小题，每小题 3 分，共 30 分）

|    |   |   |   |   |   |   |   |   |   |    |
|----|---|---|---|---|---|---|---|---|---|----|
| 题号 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 答案 | D | B | A | B | C | B | B | C | C | A  |

二、填空题（共 6 小题，每小题 3 分，共 18 分）

11. -40

12. -1（答案不唯一）

13. 1或0 (2分+1分)

14. 2186

15. 7;  $2\sqrt{3}$  (1分+2分)

16. ①②④⑤

三、解答题（共 8 小题，共 72 分）

17. 解:解不等式①得 $x < 3$ , ..... 3分

解不等式②得 $x \geq 1$ ; ..... 3分

不等式组解集为 $1 \leq x < 3$ . ..... 2分

18. (1) 证明略; ..... 5分

(2)  $BD=AC$ . ..... 3分

19. (1) 50, ..... 2分

图略; ..... 1分

(2) 320; ..... 3分

机器人社团 ..... 1分

抽取的学生中, 选机器人社团的人数最多. .... 1分

20. (1) 证明: 连接  $OC$ .

$\because AD$  切  $\odot O$  于点  $A$ ,  $\therefore \angle DAO = 90^\circ$ , ..... 1分

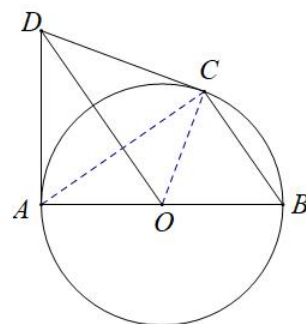
$\because BC \parallel DO$ ,  $\therefore \angle DOC = \angle OCB$ ,  $\angle DOA = \angle OBC$ ,

$\because OB = OC$ ,  $\therefore \angle OBC = \angle OCB$ ,  $\therefore \angle DOA = \angle DOC$ ,

$\because OA = OC$ ,  $OD = OD$ ,

$\therefore \triangle DOA \cong \triangle DOC$ ,  $\therefore \angle DCO = \angle DAO = 90^\circ$ , ..... 2分

$\therefore DC$  是  $\odot O$  的切线. .... 1分

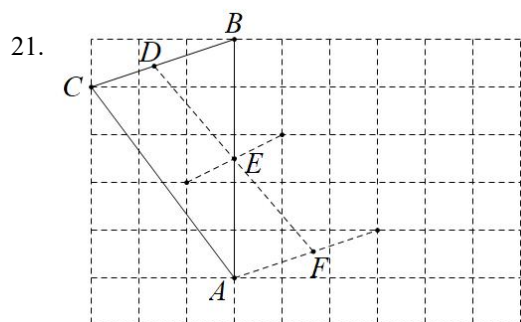


(2) 解：连接AC.

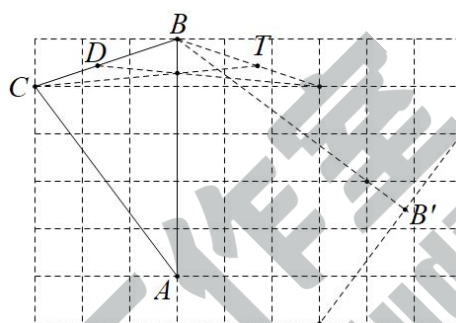
$$\because \cos \angle B = \frac{3}{5}, \therefore \cos \angle DOC = \frac{3}{5}, \frac{OC}{OD} = \frac{3}{5},$$

$$\because DC=8, \therefore OC=6, \therefore AB=12, \quad \text{..... 2分}$$

$$\because AB \text{ 是直径}, \therefore \angle ACB=90^\circ, \therefore \frac{BC}{AB} = \frac{3}{5}, \therefore BC = \frac{36}{5}. \quad \text{..... 2分}$$



(1)



(2)

22.解：(1) 对于一次函数  $y = -0.4x + 2.8$ , 令  $x=0$ , 则  $y=2.8$ ,

$\therefore P(0, 2.8)$ . 将  $P$  点坐标代入二次函数  $y = a(x-1)^2 + 3.2$  中,

$$a(0-1)^2 + 3.2 = 2.8, \therefore a = -0.4. \quad \text{..... 4分}$$

(2) 由题意, 对于一次函数  $y = -0.4x + 2.8$ , 令  $y=0$ ,  $\therefore x=7$ .

对于二次函数  $y = -0.4(x-1)^2 + 3.2$ , 令  $y=0$ , ( $x>0$ ),  $\therefore x = 2\sqrt{2} + 1. \quad \text{..... 2分}$

$$\because OA=3m, CA=2m, \therefore OC=5m. \quad \because 7-5 > 5 - (2\sqrt{2} + 1),$$

$\therefore$  武资网应选择吊球.  $\text{..... 2分}$

$$(3) b = \frac{7}{4}. \quad \text{..... 2分}$$

23. (1) 证明:  $\because \triangle BAC \sim \triangle DFE$ ,

$$\therefore \angle ACB = \angle FED, \quad \text{..... 1分}$$

$$\therefore \angle ACD = \angle AED, \quad \text{..... 1分}$$

$$\because CD = ED,$$

$$\therefore \angle DCE = \angle DEC, \quad \text{..... 1分}$$

$$\therefore \angle ACE = \angle AEC. \quad \text{..... 1分}$$

(2) 证明：过点F作FG//AC交CO延长线于点G，连接AG.

$$\therefore \angle ACE = \angle G,$$

$$\because \triangle BAC \sim \triangle DFE, BC = CD = ED,$$

$$\therefore \triangle BAC \cong \triangle DFE,$$

$$\therefore AC = FE, \quad \dots\dots\dots 1\text{分}$$

同(1)得  $\angle ACE = \angle OEF$ ,

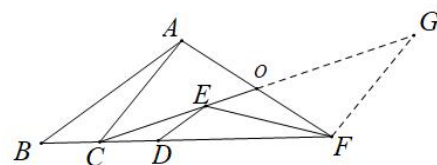
$$\therefore \angle OEF = \angle G, \therefore EF = GF = AC,$$

$$\therefore FG \parallel AC \text{ 且 } FG = AC,$$

$$\therefore \text{四边形 } ACFG \text{ 为平行四边形}, \quad \dots\dots\dots 2\text{分}$$

$$\therefore OA = OF. \quad \dots\dots\dots 1\text{分}$$

(过A作EF的平行线交CO的延长线于T，与上法一样)



$$(3) \frac{n+1-m}{mn+m} \dots\dots\dots 2\text{分}$$

24. (1)  $m = -1 \quad \dots\dots\dots 1\text{分} \quad A(2, 0); \quad \dots\dots\dots 2\text{分}$

(2)  $\because$  直线AC的解析式为  $y = -\frac{1}{2}x + 1$ ,

$$\therefore \text{设抛物线H的解析式为 } y = \frac{1}{4}(x-p)^2 - \frac{1}{2}p + 1, \quad \dots\dots\dots 1\text{分}$$

$$\text{当 } y=0 \text{ 时, } x = p \pm \sqrt{2p-4},$$

$$\therefore 2\sqrt{2p-4} = 8, \therefore p = 10, \quad \dots\dots\dots 2\text{分}$$

$$\therefore \text{抛物线H的解析式为 } y = \frac{1}{4}(x-10)^2 - 4. \quad \dots\dots\dots 1\text{分}$$

$$(3) D\left(\frac{2k-1}{k}, 0\right), \text{ 设直线DG的解析式为 } y = m\left(x - \frac{2k-1}{k}\right),$$

$$\text{令 } \frac{1}{4}x^2 - x + 1 = m\left(x - \frac{2k-1}{k}\right),$$

$\because$  直线DG与抛物线有唯一公共点G,

$$\therefore \Delta = 0, \therefore m = -\frac{1}{k}, \text{ 直线DG的解析式为 } y = -\frac{1}{k}x + \frac{2k-1}{k^2}, \quad \dots\dots\dots 3\text{分}$$

$$\therefore E\left(0, \frac{2k-1}{k^2}\right), \text{ 直线MN交y轴于点P}(0, -2k+1),$$

$$\therefore \frac{OP}{OD} = \frac{OE}{OD} = \left|\frac{1}{k}\right|,$$

$$\therefore \angle PDO = \angle DEO, \therefore \angle PDE = 90^\circ, \therefore DM \perp DG, \quad \dots\dots\dots 1\text{分}$$

$\therefore$  在射线MD上取点F，使DF=DM，在射线DG上取点T，使DT=DE，

则四边形MEFT是菱形.  $\dots\dots\dots 1\text{分}$